

See discussions, stats, and author profiles for this publication at: <http://www.researchgate.net/publication/233426583>

Time-shift Control of LLC Resonant Converters

CONFERENCE PAPER · MAY 2010

CITATION

1

DOWNLOADS

655

VIEWS

279

1 AUTHOR:



Claudio Adragna

STMicroelectronics

23 PUBLICATIONS 162 CITATIONS

SEE PROFILE

Time-shift Control of LLC Resonant Converters

Claudio Adragna, STMicroelectronics, Italy, claudio.adragna@st.com

Abstract

This paper describes a new control technique for resonant converters that outperforms the direct frequency control (DFC) method implemented in all today's commercially available control ICs. When applied to LLC resonant converters, the resulting dynamic behavior is that of a low-Q second-order system or that of a first-order system, depending on the operating conditions. Thus closing the control loop to meet stability and dynamic performance specifications becomes considerably easier. Due to its complexity, the mathematical description is limited; the approach is essentially empirical with an extensive use of simulations.

1. Introduction

Many different control methods for resonant converters have been described in the literature [1]-[9]. However, all the integrated controllers for resonant converters available in the market today use the direct frequency control (DFC) method, where the feedback signal directly controls a local oscillator, which the switching frequency of the half (or full) bridge is locked to.

Although simple, intuitive and very easily realizable in integrated form, this method is not devoid of issues. A major one is related to converter's small-signal dynamics. With reference to the LLC resonant converter (figure 1), today's most popular resonant converter, the analysis carried out in [10] shows that it is a three-pole system. Additionally, the way these poles move depends on whether the converter is operating below or above resonance and on the load. As a result, the design of a compensator that closes the control loop achieving adequate stability margin and input ripple rejection, as well as acceptable load transient response under all the specified operating conditions is not an easy task.

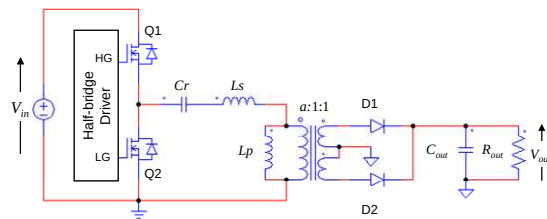


Fig. 1. LLC resonant half-bridge converter

The novel method proposed in this paper, which has been called time-shift control (TSC), alleviates this issue. In fact, simulations show that the

power stage is turned into a low-Q second-order system in some conditions or, in other cases, into a first-order system, plus the zero due to the ESR of the output capacitor. Since dominant pole approximation is allowed, the compensator design is considerably simplified.

The TSC methodology, however, does not solve other issues of the DFC method. One is the dependence of all converter variables on the input voltage, which remains essentially the same as with DFC. Another one is that both DFC and TSC are not robust against parameter variations in the resonant tank: their tolerance shifts the closed-loop operating frequency range (in the case of DFC) or time-shift range (in the case of TSC) of the converter, and any limitation imposed on the control variable must account for this.

2. Time-shift control (TSC)

2.1. The methodology

The TSC methodology consists in controlling the amount of time T_D elapsing from a zero-crossing of the tank current to the next switching of the half (or full) bridge (i.e. the zero-crossing of the first harmonic of the applied square-wave voltage) as shown in figure 2a. Note that, to achieve soft-switching for the switches Q1 and Q2, the resonant tank current must lag behind the applied square-wave voltage; so, the voltage zero-crossing commanded at the end of T_D must have the opposite sense with respect to the current zero-crossing that initiated T_D . Figure 2a shows, for example, a negative-going zero-crossing of the applied voltage (the zero reference for its ac component is $V_{in}/2$) commanded after a positive-going zero-crossing of the tank current.

It is important to note that the time-shift T_D imposed by the control circuit is included between $T_{sw}/4$ (under no-load conditions, when current is in quadrature to voltage, figure 2b) and $T_{sw}/2$ (at the boundary with capacitive-mode operation, when current and voltage are in-phase, figure 2c), being T_{sw} the switching period of the converter.

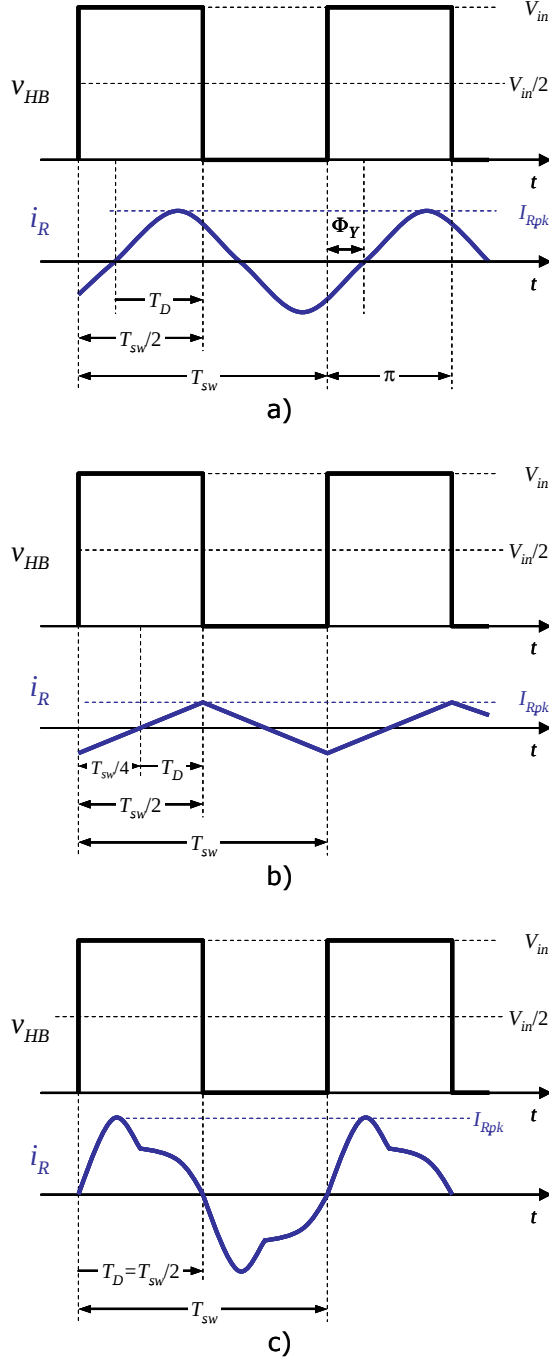


Fig. 2. a) concept of time-shift control;
b), c) range of the control variable T_D

Conceptually, with TSC an inner loop is closed and the outer loop that regulates the output voltage provides the reference for the inner loop. This inner loop is illustrated in figure 3.

The feedback variable is the tank circuit current i_R , whose zero-crossings are used as the starting point for the time-shift, while the input quantity to the tank circuit is the applied square-wave voltage v_{HB} . In the frequency domain, therefore, the tank circuit is represented by its input admittance $Y_i(j\omega)$. The block $e^{-j\omega T_D}$ represents the time-shift imposed by the control circuit, while the block designated with $D(I_{Rpk1})$ represents the inverter action of the half (or full) bridge.

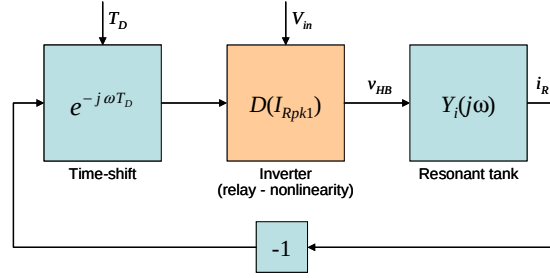


Fig. 3. Inner loop that controls the time-shift between tank current and input voltage zero-crossings

T_D is the reference for the loop, while converter's input voltage V_{in} can be seen as a disturbance.

Since the tank circuit responds primarily to the first harmonic of v_{HB} and negligibly to the higher order harmonics, this loop can be analyzed using the describing function approach [11]. With this method, the non-linear part of the system, i.e. the inverter action of the half (or full) bridge, which in control system terminology is a relay, is represented in the frequency domain by its describing function $D(I_{Rpk1})$, where I_{Rpk1} is the peak value of the first harmonic of i_R . In our case, since the ac voltage applied to the resonant tank oscillates between $\pm V_{in}/2$, it is:

$$D(I_{Rpk1}) = \frac{2V_{in}}{\pi I_{Rpk1}}. \quad (1)$$

According to the describing function method, the intersection point of the loci of the negative inverse of the describing function (1) and of the transfer function of the linear blocks in the loop ($Y_i(j\omega)e^{-j\omega T_D}$ in our case) defines a possible limit cycle. The values of I_{Rpk1} and of the angular frequency ω at the intersection point define the amplitude and the frequency of the oscillation, respectively, provided the limit cycle is stable.

Stability can be easily assessed: if the points near the intersection and along the increasing I_{Rpk1}

side of the curve $-1/D(I_{Rp1})$ are not encircled by the locus of the linear part, then the corresponding limit cycle is stable.

Figure 4 shows the Nyquist plots of $-1/D(I_{Rp1})$ and $\tilde{Y}_i(j\omega)e^{-j\omega T_D}$, where $\tilde{Y}_i(j\omega)$ is the tank circuit's input admittance normalized to the tank circuit's characteristic admittance $Y_R = (Cr/Ls)^{1/2}$. Due to the delay term $e^{-j\omega T_D}$, there are actually infinite intersections and so possible limit cycles; however, considering that $T_{sw}/4 \leq T_D \leq T_{sw}/2$, the only significant intersection is the one falling in the ω range $(\pi/2T_D, \pi/T_D)$ shown in figure 4. The corresponding limit cycle, according to the above mentioned criterion, is stable, and this holds true regardless of T_D (provided it is in the correct range) and of the parameters in $\tilde{Y}_i(j\omega)$.

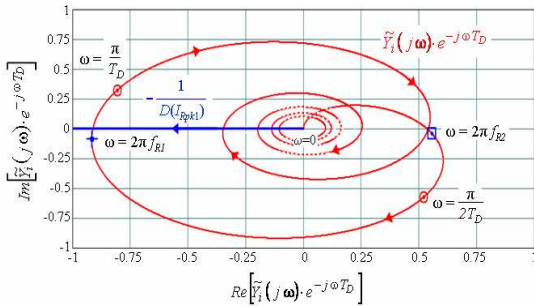


Fig. 4. Nyquist plots of the transfer function of the linear part and the negative inverse describing function of the inverter used to study the oscillations of the loop in figure 3.

Therefore, TSC features the same property of absolute stability as the self-sustained oscillation (SSO) technique illustrated in [7]-[8]; unlike TSC, SSO controls the phase-shift between the resonant tank current and the applied square-wave voltage, so that it enjoys control robustness as well. However, the implementation of the modulator is relatively complex. The TSC modulator, instead, is obtained with a simple modification in the operation of the oscillator included in a standard control IC.

2.2. TSC modulator implementation

Figure 5 shows the block diagram of the oscillator in a commercially available resonant controller IC, the L6599A, which implements DFC.

A capacitor, C_T , is alternately charged and discharged with the same current I_F whenever the voltage across C_T falls below a lower threshold V_v and exceeds an upper threshold V_p , respectively. In this way, a series of contiguous ascending and descending ramps is generated, which results in a symmetrical triangular waveform across C_T .

The driving signals for Q1 and Q2, HG and LG, are locked to this triangular wave as shown in the timing diagram in figure 5. The dead-times between the HG and LG pulses are neglected because not significant in this context.

Modulating the current I_F changes the ramp slope and, then, the oscillation frequency f_{sw} , thus performing DFC.

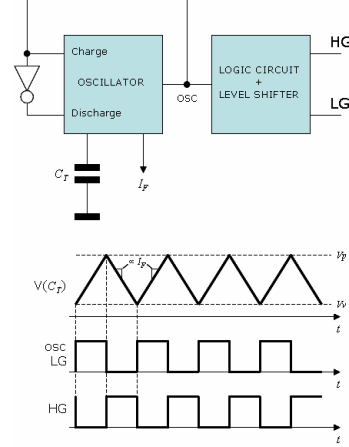


Fig. 5. Block diagram and key waveforms of the oscillator of the L6599A (DFC)

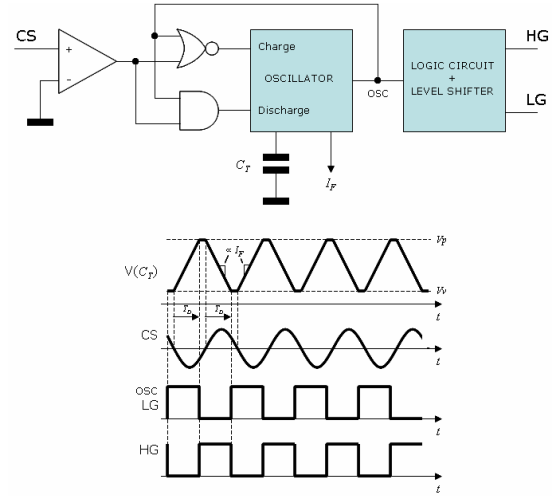


Fig. 6. Block diagram and key waveforms of a TSC modulator derived from the oscillator in figure 5.

Figure 6 shows how the oscillator of figure 5 can be modified to perform TSC. A comparator has been added that senses the CS signal, voltage image of the tank current i_R (derived, for example, with a resistor connected in series to it). This comparator, along with the NOR gate, enables the charge of C_T when CS (i.e. i_R) is negative and, through the AND gate, its discharge when CS (i_R) is positive.

In this way, each ramp across C_T starts after a zero crossing of CS (i_R) and lasts until it crosses either V_v or V_p , which causes the half (or full) bridge to be switched. Afterwards, the voltage on C_T stays flat until the next zero-crossing of CS (i_R), when another ramp starts. With this mechanism, the duration of a ramp does not determine half the switching period $T_{sw}/2$ and, ultimately, the switching frequency f_{sw} , as with the modulator in figure 5. It determines the distance in time T_D between a zero-crossing of the tank current and the next switching of the half (full) bridge:

$$T_D = C_T \frac{V_p - V_v}{I_F}. \quad (2)$$

In this case, modulating I_F modifies this distance in time, thus performing TSC. The consequent change in frequency is not directly controlled and depends on the phase-shift between i_R and v_{HB} .

From (2) it is apparent that the TSC modulator is a non-linear block, thereby its gain will change with the operating point (i.e. with I_F). However, it does not introduce frequency dependence, at least in the frequency range that is interesting for the design of a compensator. The small-signal gain is found by differentiating (2):

$$G_{Mod} = \frac{\hat{T}_D}{\hat{i}_F} = -C_T \frac{V_p - V_v}{I_F^2}, \quad (3)$$

where $\hat{x}$ denotes the signal component of x .

3. TSC dynamic behavior

3.1. Theoretical considerations

The block diagram of the feedback loop that controls the output voltage of an LLC resonant converter operated with TSC is shown in figure 7.

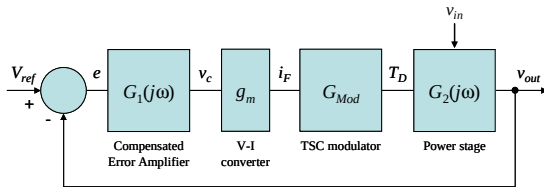


Fig. 7. Block diagram of the overall control loop

$G_2(j\omega)$ is the small-signal transfer function of the power stage, i.e. of the closed-loop control system illustrated in figure 3. Its T_D reference value is provided by the TSC modulator, whose transfer function G_{Mod} is given by (3).

The modulator is current controlled and a transconductance block g_m converts the control volt-

age v_c (error amplifier output) to the controlling current i_F .

$G_1(j\omega)$ is the transfer function of the compensated error amplifier. Once all the other blocks in the loop are known, $G_1(j\omega)$ is designed to meet a design target normally given as phase margin and crossover frequency of the open-loop system.

$G_2(j\omega)$ can be expressed as follows:

$$G_2(j\omega) = \frac{\hat{v}_{out}}{\hat{T}_D} = \frac{\hat{v}_{out}}{\hat{i}_F} \frac{\hat{f}_{sw}}{\hat{\Phi}_Z} \frac{\hat{\Phi}_Z}{\hat{T}_D}. \quad (4)$$

Note that the first factor on the right-hand side of (4) is the small-signal transfer function of the power stage with DFC.

The term Φ_Z in the other factors is the phase-shift between the tank current and the applied square-wave voltage v_{HB} . In the spirit of the “first-harmonic approximation” approach, as already done in section 2.1 with the describing function method, Φ_Z is the phase of the input impedance $Z_i(j\omega) = 1 / Y_i(j\omega)$ of the tank circuit. The second factor is, therefore, the reciprocal of the small-signal expression of $\angle Z_i(j\omega)$, again with DFC.

The third factor relates phase-shift to time-shift. It can be calculated with the aid of figure 2a:

$$\Phi_Z = \pi \frac{T_s/2 - T_D}{T_s/2} \rightarrow \frac{\hat{\Phi}_Z}{\hat{T}_D} = -2\pi f_{sw} = -\omega. \quad (5)$$

3.2. Empirical approach

As can be inferred from (4), the analytical derivation of $G_2(j\omega)$ is a challenging task. For this reason, an empirical approach has been preferred in this context: the behavior is characterized by means of a number of simulations carried out on an exemplary LLC resonant converter in both the time and the frequency domains.

The key electrical parameters of the exemplary LLC resonant converter are listed in Table 1.

In the time-domain simulations a small step-change $\Delta V_c \ll V_c$ is given to the steady-state value V_c and the resulting change ΔV_{out} in V_{out} is observed. The amplitude of ΔV_{out} and the 10-90% rise/fall times of the response are noted. To minimize the effect of the output voltage ripple on the measurement of ΔV_{out} , the reading is done via two cascaded second-order filters (40 kHz cut-off frequency and 0.7 damping ratio each).

In the frequency-domain simulations, the test frequency is swept from 10 Hz to 30 kHz.

All simulations are carried out at the extremes of the input voltage range and at the nominal input voltage (where the converter runs close to resonance) with 100%, 50% and 10% load.

Parameter	Symbol	Value	Unit
Dc input voltage range	$V_{in,min} - V_{in,max}$	300 - 440	V
Nominal dc input voltage	$V_{in,nom}$	400	V
Regulated output voltage	V_{out}	12	V
Maximum output power	$P_{out,max}$	250	W
Transformer turn-ratio	a	16:1	---
Series resonant inductor	L_s	105	μ H
Parallel resonant inductor	L_p	560	μ H
Resonant capacitor	C_r	39	nF
Output capacitor	C_{out}	8 x 470	μ F
Output capacitor's ESR	R_C	38/8	m Ω
HB node parasitic capacitance	C_{HB}	180	pF
Dead-time of driver circuit	T_d	300	ns
Q1 and Q2 on-resistance	$R_{DS(on)}$	0.35	Ω
Secondary diode forward drop	V_F	0.5	V
Oscillator capacitor	C_T	330	pF
Oscillator voltage swing	$V_p - V_v$	3	V
V-I block transconductance	g_m	233	μ S

Table 1. Key parameters of the exemplary LLC resonant HB converter used in the simulations

The time domain simulations, shown in the diagrams of figures 8 to 10, point out an open-loop response that is always overdamped.

However, a close look at the step response with 100% load and $V_{in} = V_{in,min}$ shows that the rise and fall times of V_{out} are not compatible with a single time constant. At lower load and/or higher input voltage, instead, the step response fits an exponential curve very well.

This suggests that $G_2(j\omega)$ might have two real poles, one of which tends to move to high frequency at high input voltage and/or light load.

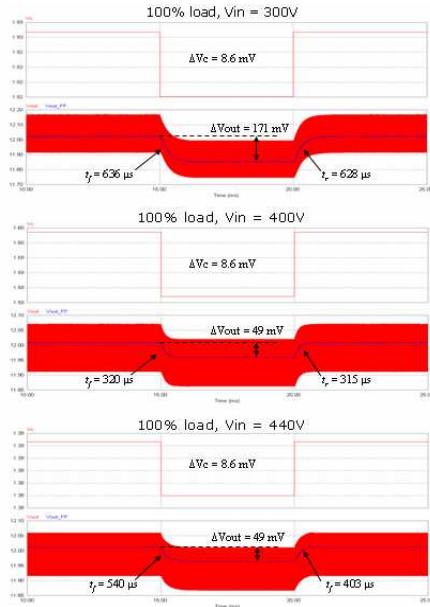


Fig. 8. Time-domain simulations at 100% load with $V_{in} = 300, 400, 440$ V; upper traces: V_c ; lower traces: V_{out} (red), filtered V_{out} (blue)

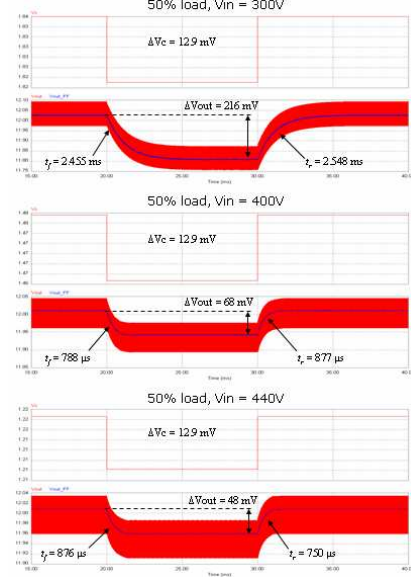


Fig. 9. Time-domain simulations at 50% load with $V_{in} = 300, 400, 440$ V; same traces as in figure 8

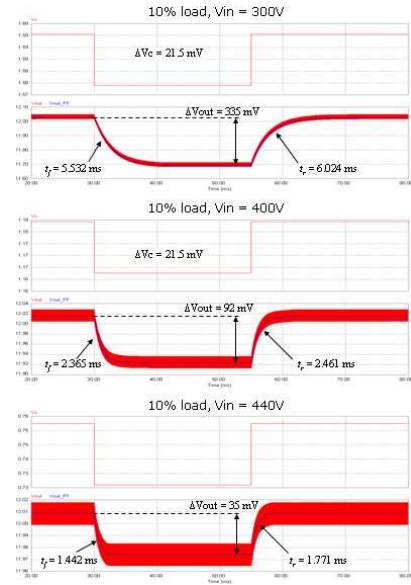


Fig. 10. Time-domain simulations at 10% load with $V_{in} = 300, 400, 440$ V; same traces as in figure 8

This is confirmed by the results of the frequency-domain simulations in figure 11, where gain and phase lag associated to $\hat{v}_{out}/\hat{v}_c$ are shown.

The Bode plots show clearly two real poles at 100% load and $V_{in} = V_{in,min}$. However, with respect to a single pole plot, the extra phase lag at the 3-dB cutoff frequency is only $\approx 6^\circ$. This means that the two poles are separated by about a decade and that using dominant pole approximation is definitely possible at frequencies below cross-over (low-Q second-order system).

At lower load and/or higher input voltage, the Bode plots become essentially those of a first-order system.

In all Bode plots, besides the zero due to the ESR of C_{out} , an RHP zero seems to lie beyond ≈ 20 kHz: gain slope tends to flatten out or even rise, while phase lag further increases.

Still at 100% load and $V_{in} = V_{in,min}$, crossover occurs with the lowest phase margin ($\approx 40^\circ$). Therefore, designing the compensator for closed loop stability in these conditions ensures closed-loop stability under all conditions.

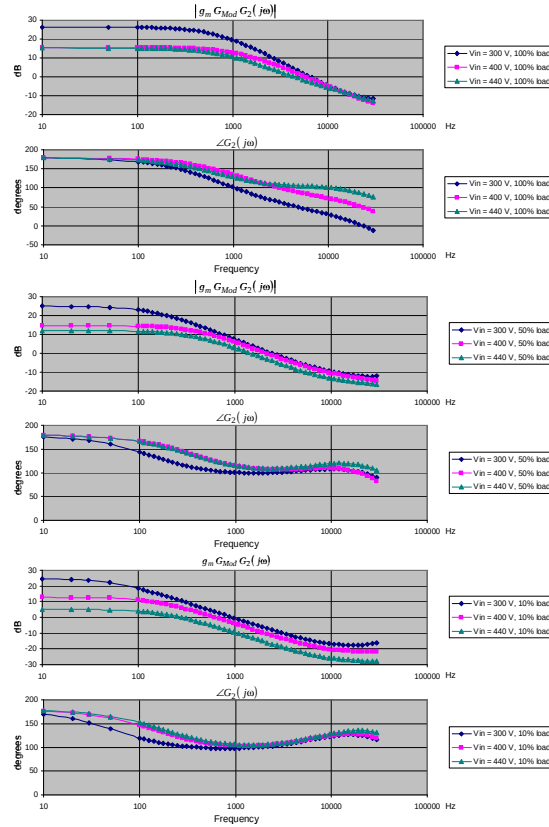


Fig. 11. Frequency-domain simulations at 100%, 50% and 10% load with $V_{in} = 300, 400, 440$ V

3.3. Future work

The previous analysis is far from being exhaustive. Further investigation should:

- 1) relate dynamic properties to the parameters of the resonant tank (L_s, L_p, C_r, a);
- 2) ascertain if poles are always real or may become conjugate complex;
- 3) ascertain if an RHP zero is really present and is always confined to high frequency.

It goes without saying that finding an analytical representation of $G_2(j\omega)$ would be most desirable.

4. Summary

A novel control technique for resonant converters, the Time-shift control (TSC), has been presented and its application to LLC resonant converters studied. Large signal behavior and stability have been analyzed with the describing function method. Its dynamic behavior has been characterized by simulations with PSIM and found to be either that of a low-Q second-order system or that of a first-order system. Dominant pole approximation can be used as far as the design of a compensator is concerned.

A possible TSC modulator, derived from the oscillator of a commercially available control IC for resonant converters has been presented.

Although not solving all the issues related to the presently most used control technique (Direct frequency control, DFC), the TSC method offers significant improvements to stability.

5. Literature

- [1] R. Oruganti, F.C. Lee: "Resonant Power Processors, Part II – Methods of Control", IEEE Trans. on Industry Applications, Vol. IA-21, N. 6, Nov/Dec 1985
- [2] G.B. Joong, C.T. Rim, G.H. Cho: "Integral cycle control of the series resonant converter", IEEE Trans. on Power Electronics, Vol. 4 N. 1, pp 83-91, Jan 1989
- [3] M.G. Kim, M.J. Youn: "An Energy Feedback Control of Series Resonant Converter", IEEE Trans. on Power Electronics, Vol. 6 N. 3, pp 338-345, Jul 1991
- [4] W. Tang, C.S. Leu, F.C. Lee: "Charge Control for zero-voltage-switching multiresonant converter", IEEE PESC, 1993 Rec. pp. 229-233
- [5] J. Hong, D. Maksimović, R.W. Erickson, I. Khan, "Half-Cycle Control of the Parallel Resonant Converter Operated as a High Power Factor Rectifier," IEEE Trans. on Power Electronics, Vol. 10, N. 1, pp. 1-8, Jan 1995.
- [6] L. Rossetto: "A simple control technique for series resonant converters", IEEE Trans. Power Electronics, Vol. 11 N. 4, pp 554-559, Jul 1996
- [7] H. Pinheiro, P.K. Jain, G. Joós: "Self-sustained Oscillating Resonant Converters Operating Above the Resonant Frequency", IEEE Trans. on Power Electronics, Vol. 14 N. 5, pp 803-814, Sept 1999
- [8] M.Z. Youssef, H. Pinheiro, P.K. Jain: "Self-sustained Phase-Shift Modulated Resonant Converters: Modeling, Design and Performance", IEEE Trans. on Power Electronics, Vol. 21 N. 2, pp 401-414, Mar 2006
- [9] Y. Lu, K.W.E. Cheng, S.L. Ho: "Quasi Current Mode Control for the Phase-Shifted Series Resonant Converter", IEEE Trans. on Power Electronics, Vol. 23, N. 1, Jan 2008
- [10] B. Yang: "Topology Investigation for Front End Dc-dc Power Conversion for Distributed Power System", PhD Dissertation, VPEC, Chap. 6, pp. 187-236
- [11] J.J.E. Slotine, W. Li: "Applied Nonlinear Control", Englewood Cliffs, NJ; Prentice-Hall 1991, pp. 159-190.